

Equal face-coloring counts with opposite matching parities

Disproof of Conjecture 00000003477

ziangni-sys

10 October 2026

Result. The planar trees K_2 and $K_{1,3}$ have identical face-coloring class counts but opposite perfect-matching parities. Face-coloring class count therefore does not uniquely determine the parity asserted in the conjecture.

The source and the two actual graphs

Both language versions define the parity through the number of perfect matchings of a planar graph, and assert that this parity is uniquely decided by the class count of face colorings. We disprove that first assertion. No claim about the later nonplanar or Petersen-class assertions is needed.

Let $G_1 = K_2$ have vertices $0, 1$ and its single edge. Let $G_2 = K_{1,3}$ have vertices $0, 1, 2, 3$, center 0 , and edges $01, 02, 03$. Both have an even number of vertices. Both are connected planar trees: the first is a line segment and the second is a star drawn with three nonintersecting radial segments.

There is exactly one perfect matching of G_1 , its single edge. There is no perfect matching of G_2 : each leaf can only be matched to the center, while a matching can use the center only once. Equivalently, a perfect matching is a fixed-point-free involution of the actual vertices that sends every vertex along a graph edge. In the star it would send all three leaves to 0 , contradicting injectivity. Thus

$$M(G_1) = 1, \quad M(G_2) = 0, \quad M(G_1) \bmod 2 \neq M(G_2) \bmod 2.$$

The actual planar faces

For completeness, use the standard rotation-system description of the explicit plane embeddings. A dart is an oriented actual edge. The face successor is the local rotation after dart reversal.

For G_1 , enumerate darts as $(0, 1), (1, 0)$. The local rotations are the identity because each vertex has degree one. Dart reversal interchanges the two darts, giving one face orbit.

For G_2 , enumerate darts in order

$$(0, 1), (1, 0), (0, 2), (2, 0), (0, 3), (3, 0).$$

Choose the local cyclic order $1, 2, 3$ at the center; the rotations at leaves are trivial. On dart indices $0, \dots, 5$, reversal is $(0\ 1)(2\ 3)(4\ 5)$, while the rotation is $(0\ 2\ 4)$ and fixes $1, 3, 5$. Their composition has the single face cycle

$$(0\ 1\ 2\ 3\ 4\ 5).$$

These are actual dart operations: reversal swaps the endpoints, and rotation preserves the source vertex. The connected embeddings have $V - E + F = 2$ in both cases, agreeing with their explicit planar drawings. Each graph therefore has exactly one face, the unbounded face in the plane.

The same coloring classes do not determine parity

Under the standard convention that face adjacency requires two distinct faces sharing an edge, both face-adjacency graphs are the one-vertex graph with no edges. For a palette of q colors, each has exactly q labeled face colorings. If colorings are identified under permutations of the palette, each has exactly one coloring class for $q \geq 1$ and none for $q = 0$. The bijection between their singleton face sets induces an equivariant bijection of their coloring sets, so it also identifies these class counts.

A convention that forbids the same color on both sides of a bridge treats a face as adjacent to itself. Then both graphs admit no proper face coloring and again have the same class count, zero. Thus the bridge convention cannot distinguish the graphs. A convention restricted only to bridgeless graphs would impose an additional domain restriction absent from the source.

The input face-coloring class counts agree, but the asserted output matching parities disagree. Therefore no unique decision of parity from those counts is possible, disproving the first conjunct.

Formal verification

Lean defines both actual simple graphs, counts respecting-edge fixed-point-free involutions, and proves the perfect-matching counts 1 and 0. It explicitly enumerates the actual dart sets and verifies those enumerations are bijective. It verifies dart reversal, local rotations, and the face-successor rule, computes each face orbit, and constructs the quotient face types. Both are singleton types. Their face-coloring counts agree for every palette size, and a coloring equivalence identifies their colorings; the matching parities differ. Finite decisions use Lean's checked kernel reduction, not unchecked native decisions.

Run `lake build` in `lean/`. The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`

Printed final theorem audits contain only standard axioms. No admitted results, custom axioms, or unsafe code are used.